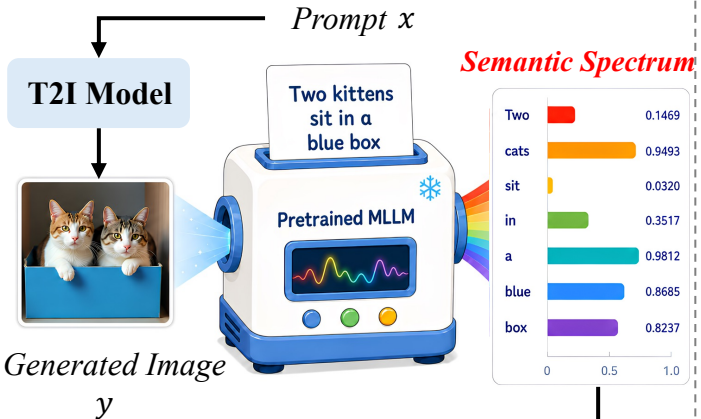


(a) SpectraReward from Semantic Spectrum



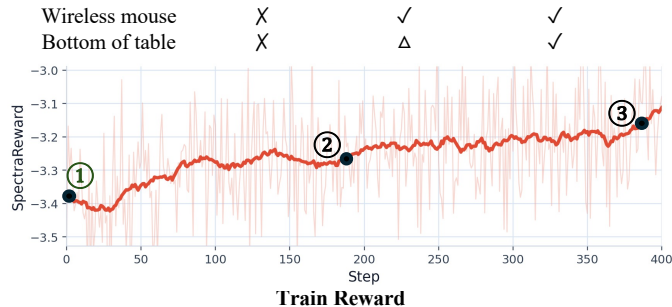
$$\text{SpectraReward} = \frac{1}{T-1} \sum_{t=1}^{T-1} \log P_M(x_{t+1} | x_{\leq t}, y)$$

(b) Training Dynamics

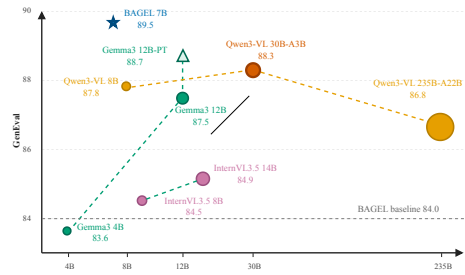
Prompt: *A wireless mouse on the bottom of the table is smaller than a crimson scarf basking in sunlight on the top.*



① Early ② Mid ③ Late



(c) Reward MLLM Comparison



(d) Quantitative Comparison

